Artificial Intelligence (AI2002)
Quiz # 01

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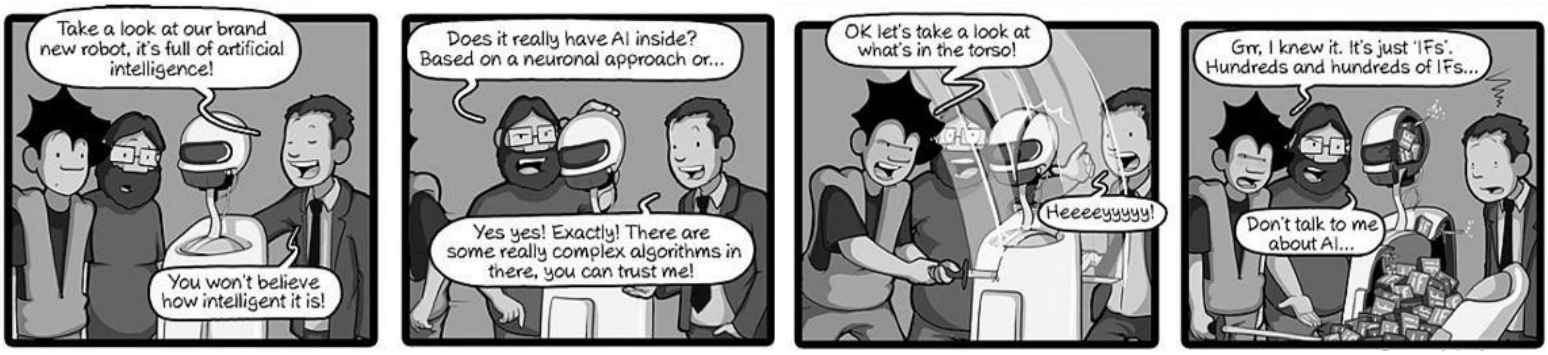
Roll No	**Section**	**Student Signature**

Q1. Consider the comic below and answer the questions that follow: - [8]



A. Which dimension of AI definitions best matches the context in the above comic? Give strong reasons.
B. What type of Agent did the robot turn out to be? Why were the guys disappointed to find many IFs? Isn't it AI?.
C. Give PEAS for solving a jigsaw puzzle.
D. Which of the categories of Nature of environments does solving jigsaw puzzles belong to? Cover all the characteristics of the task environment. (**HINT**: fully vs partially observable, deterministic vs nondeterministic etc.)

CLO2: To identify and relate methods of search and practically apply the corresponding techniques.

Q2. Suppose you have a transportation network with 10 cities represented by the adjacency matrix below. Each connection between cities has a specific weight (representing distance in kilometers). The goal is to analyze paths between City A and City J using different search algorithms. [7]

	A	B	C	D	E	F	G	H	I	J
A	0	2	4	0	0	0	0	0	0	0
B	2	0	0	7	0	3	0	0	0	0
C	4	0	0	1	5	0	0	0	0	0
D	0	7	1	0	0	0	2	6	0	0
E	0	0	5	0	0	0	0	0	4	0
F	0	3	0	0	0	0	8	0	0	5
G	0	0	0	2	0	8	0	0	0	3
H	0	0	0	6	0	0	0	0	1	0
I	0	0	0	0	4	0	0	1	0	2
J	0	0	0	0	0	5	3	0	2	0

a. Using BFS, what is the shortest path from City A to City J (considering unweighted paths)?
b. Using DFS, find the longest path from City A to City J without revisiting any city.
c. Using the Greedy Algorithm (always choose the nearest next city), what path would you find from City A to City J? Is this the optimal path? Justify your answer.